Minilab - Git

1. Configuration

```
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git config --global user.name "Ayzeen Ashraf"
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git config --global user.name aashraf@scu.edu
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git config --global init.defaultBranch main
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 %
```

2. git init and git status

```
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % ls
file1.txt       file2.txt       file3.txt       hello.cpp       myclass.cpp     myclass.h
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % ls -a
.               ..              .git            file1.txt       file2.txt       file3.txt       hello.cpp       myclass.cpp     myclass.h
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file1.txt
        file2.txt
        file3.txt
        hello.cpp
        myclass.cpp
        myclass.h

nothing added to commit but untracked files present (use "git add" to track)
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 %
```

3. git add, git commit, git log

```
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git add hello.cpp
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   hello.cpp

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file1.txt
        file2.txt
        file3.txt
        myclass.cpp
        myclass.h

(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git commit -m "First commit"
[main (root-commit) 29f490a] First commit
 Committer: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 11 insertions(+)
 create mode 100644 hello.cpp
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git log
commit 29f490a2262bac8f45bd20c7dba934def48f41fd (HEAD -> main)
Author: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Date:   Mon Feb 9 14:52:56 2026 -0800

    First commit
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 %
```

4. More git add

```
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file1.txt
        file2.txt
        file3.txt
        myclass.cpp
        myclass.h

nothing added to commit but untracked files present (use "git add" to track)
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   hello.cpp

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file1.txt
        file2.txt
        file3.txt
        myclass.cpp
        myclass.h

no changes added to commit (use "git add" and/or "git commit -a")
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git add hello.cpp
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git commit -m "Second commit"
[main aa3c9b8] Second commit
 Committer: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 4 insertions(+)
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git log
commit aa3c9b81119358fce9c95ed7b025f85c3d76ff67 (HEAD -> main)
Author: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Date:   Mon Feb 9 18:04:01 2026 -0800

    Second commit

commit 29f490a2262bac8f45bd20c7dba934def48f41fd
Author: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Date:   Mon Feb 9 14:52:56 2026 -0800

    First commit
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 %
```

5. gitignore, more git add

.gitignore

```
myclass.h
*.txt
# ignored files
```

```
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git add .
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git commit -m "Add all files"
[main 1f547eb] Add all files
 Committer: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 2 files changed, 15 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 myclass.cpp
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 %
```

6. git diff

```
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git diff
diff --git a/hello.cpp b/hello.cpp
index 2058303..b1fe13a 100644
--- a/hello.cpp
+++ b/hello.cpp
@@ -9,7 +9,10 @@ int main () {




+
+

   MyClass a(5);
   cout << a.getX() << endl;
+
 }
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 %
```

7. More git add

```
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git commit -a -m "description"
[main dd5f4ae] description
 Committer: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 2 files changed, 4 insertions(+), 2 deletions(-)
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 %
```

8. Branches

```
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git branch mybranch
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git checkout mybranch
Switched to branch 'mybranch'
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git add hello.cpp
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git commit -m "Third commit"
[mybranch 1e895da] Third commit
 Committer: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 2 insertions(+)
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git checkout main
Switched to branch 'main'
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git log
commit dd5f4aea6bac6f7f02fb71fdf0ffe668d6f2fca0 (HEAD -> main)
Author: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Date:   Mon Feb 9 18:16:22 2026 -0800

    description

commit 1f547eb76f05c08ada3dd2da6bdb267b9ff20331
Author: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Date:   Mon Feb 9 18:12:15 2026 -0800

    Add all files

commit aa3c9b81119358fce9c95ed7b025f85c3d76ff67
Author: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Date:   Mon Feb 9 18:04:01 2026 -0800

    Second commit

commit 29f490a2262bac8f45bd20c7dba934def48f41fd
Author: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Date:   Mon Feb 9 14:52:56 2026 -0800

    First commit
```

```
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git merge -m "message" mybranch
Updating dd5f4ae..1e895da
Fast-forward (no commit created; -m option ignored)
 hello.cpp | 2 ++
 1 file changed, 2 insertions(+)
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git log
commit 1e895dadbdd21ece97e801cf5fa141ad3195d101 (HEAD -> main, mybranch)
Author: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Date:   Mon Feb 9 18:18:31 2026 -0800

    Third commit

commit dd5f4aea6bac6f7f02fb71fdf0ffe668d6f2fca0
Author: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Date:   Mon Feb 9 18:16:22 2026 -0800

    description

commit 1f547eb76f05c08ada3dd2da6bdb267b9ff20331
Author: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Date:   Mon Feb 9 18:12:15 2026 -0800

    Add all files

commit aa3c9b81119358fce9c95ed7b025f85c3d76ff67
Author: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Date:   Mon Feb 9 18:04:01 2026 -0800

    Second commit

commit 29f490a2262bac8f45bd20c7dba934def48f41fd
Author: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Date:   Mon Feb 9 14:52:56 2026 -0800

    First commit
```

```
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git add hello.cpp
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git commit -m "Fourth commit"
[main 096dd62] Fourth commit
 Committer: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 2 insertions(+)
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git checkout mybranch
Switched to branch 'mybranch'
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git checkout main
Switched to branch 'main'
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git checkout mybranch
Switched to branch 'mybranch'
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git add hello.cpp
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git commit -m "Fifth commit"
[mybranch 615873d] Fifth commit
 Committer: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 1 insertion(+), 1 deletion(-)
```

```
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git checkout main
Switched to branch 'main'
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git merge -m "message" mybranch
Auto-merging hello.cpp
CONFLICT (content): Merge conflict in hello.cpp
Automatic merge failed; fix conflicts and then commit the result.
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git add hello.cpp
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git commit -m "After fixing"
[main 0e6048e] After fixing
 Committer: aashraf@scu.edu <ayzeenashraf@AyzeensMacBookAir-254.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git merge -m "message" mybranch
Already up to date.
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 % git branch -d mybranch
Deleted branch mybranch (was 615873d).
(base) ayzeenashraf@AyzeensMacBookAir-254 minilab3 %
```